Skyler Grandel

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EDUCATION

Vanderbilt University **Nashville, TN**

Bachelor of Science in Computer Science *Graduation: May 2022*

- **GPA:** 3.85
- **Research Mentors:** Douglas C. Schmidt, Jesse Spencer-Smith

Doctor of Philosophy in Computer Science *Expected Graduation: Summer 2026*

- **GPA:** 4.0
- **Advisor:** Kevin Leach

Research Interests: AI for Software Engineering, Human Factors in Software Engineering, Human Factors in Cybersecurity, LLMs and Generative AI, Software Engineering Education and Training.

PUBLICATIONS

COMCAT: Leveraging Human Judgment to Improve Automatic Documentation and Summarization.
Grandel, S., Andersen, S. T., Huang, Y., & Leach, K. (2025).
In the *ACM Transactions on Software Engineering and Methodology*.

- https://dl.acm.org/doi/abs/10.1145/3742475

Applying Large Language Models to Enhance the Assessment of Parallel Functional Programming Assignments.
Grandel, S., Schmidt, D. C., & Leach, K. (2024).
In *Proceedings of the 2024 International Workshop on Large Language Models for Code (pp. 1-9).*

- Awarded Best Presentation at the 2024 International Workshop on Large Language Models for Code. Please see https://llm4code.github.io/papers/.
- https://dl.acm.org/doi/abs/10.1145/3643795.3648375

Applying Large Language Models to Enhance the Assessment of Java Programming Assignments.
Grandel, S., Schmidt, D. C., & Leach, K. (2025).
In Proceedings *of the 33rd ACM International Conference on the Foundations of Software Engineering. (FSE25 SEET)*

- https://dl.acm.org/doi/abs/10.1145/3696630.3727236

A Human Study of Cognitive Biases in Web Application Security.
Yang, Y., Grandel, S., Huang, Y., & Leach, K. (2024).
arXiv preprint arXiv:2505.12018 (2025). Submitted to the *34th USENIX Security Symposium*

- https://arxiv.org/abs/2505.12018

A Human Study of Automatically Generated Decompiler Annotations.
Yang, Y., Grandel, S., Lacomis, J., Schwartz, E., Vasilescu, B., Le Goues, C., & Leach, K. (2025).
In the *55th Annual IEEE/IFIP International Conference on Dependable Systems and Networks (DSN)*.

- https://doi.org/10.1109/DSN64029.2025.00026

PROFESSIONAL EXPERIENCES

Vanderbilt University Medical Center **Nashville, TN**

Audio-Visual Assistant *Aug. 2018 – May 2020*

- Solved hardware and software issues for the Vanderbilt School of Medicine, working with advanced equipment and multiple operating systems.
- Prepared rooms on a tight schedule for classes, presentations, and other technologically assisted uses.
- Fixed network problems for students, medical doctors, and university guests; aiding less technologically adept users.

Car Tech LLC **Opelika, AL**

Information Technology and Cybersecurity Intern *May. 2021 – Aug 2021*

- Solved hardware and software issues for Car Tech's various divisions, setting up equipment, fixing security devices, and managing the company's server.
- Researched, wrote, revised, presented, and implemented information security documents to follow ISO 27001 guidelines to ensure state-of-the-art standards for business security were met.
- Researched market data to successfully renegotiate a contract with an external company, resulting in significant monthly cost savings of up to $8000.

Vanderbilt University School of Engineering **Nashville, TN**

Teaching Assistant and Grader *Aug. 2020 - Present*

- Instructed students in Compilers, Web Systems, Digital Systems, Operating Systems, Parallel Functional Programming, Concurrent Object-Oriented Programming, Scalable Microservices, and Software Engineering courses.
- Developed assignments, exams, and lectures; Regularly lectured for undergraduate and graduate courses.
- Assisted professors in grading dozens of assignments, tests, etc. in a timely manner for up to 19 hours per week.

EXTRACURRICULAR EXPERIENCES

Deelzebub (Information Link: https://ifdb.org/viewgame?id=tx41ishkacmx8lij) **Remote**
Lead Programmer *May 2020 – Sept 2020*

- Committed 4 months to program an interactive fiction, text-based, online parser game, incorporating complex story paths and player interaction with an in-game world and the characters and objects it contains
- Designed from the bottom up, programming user interactions and logic for a total of over 15,000 lines of code
- Self-taught TADS 3, the programming language used to create the game
- Bridged creative differences within a 4-person team in order to produce a professional, standalone product on a strict deadline

Research: Love In A Big World Project **Remote**
Undergraduate Researcher *Aug 2021 – May 2023*

- Research to develop a mental health application for middle and high school aged students struggling to cope with isolation and lack of access to mental health services during the COVID-19 Pandemic.
- The project utilizes machine learning and natural language processing to recommend content for users without access to user history or invasion of privacy.
- Developed server with Spring, utilizing sophisticated development and design patterns and parallel computing techniques.
- Beginning in July 2022, I worked part-time for the associated company, BlueWonder Creative, as a data scientist to aid in data preparation for training and testing AI models used in this application. This is not included in my work experience for brevity.

SKILLS

- **Software Development Skills**: Software Pattern and Algorithm Use and Integration, Programming Language Analysis and Selection, Database Design and Management, Operating System Kernel Driver Development (Linux), Object-Oriented, Logic, and Functional Programming Development, Scripting, Website Development, Concurrency and Multithreading in Java and POSIX libraries, Server Development with Spring Framework, Machine Learning and NLP, Cybersecurity, Reverse Engineering
- **Computer Language Skills**: Java, Python, C, C++, C#, JavaScript, HTML/CSS, SQL, Assembly, Racket, Prolog, Spring, PHP